14.4 OVERVIEW OF PARTIAL DERIVATIVES & HIGHER ORDER PARTIAL DERIVATIVES

In this section, we just summarise basic concepts, notations and related issues discussed in Unit11. Most of the explanations are in terms of relevant examples.

For our purpose, a dependent variable may be specified in terms of independent variables, on which it depends, mainly in the following THREE forms:

i) ***Explicitly***, as a single function of independent variables

ii) ***Implicitly***, in the form an equation, in which, neither side of '=' being a single variable

iii) As a ***composition***, or a non-circular chain of compositions of functional definitions of some variables in terms of other variables.

In this regard, we may recall the following facts

i) In the case of functions, say z= f(x), of ***only one*** variable, the derivative, $(dz/dx)_{x=a}$ or $f'(a)$ represents the ***rate of change*** of the function f(x), ***at the point x=a,*** as the independent variable x changes in the neighbourhood of x =a,

ii) the issue of partial derivative of a variable arises only when a variable is dependent, directly or indirectly, on two or more, variables.

   Thus, our concerns in the case of functions of two or more variables, are to discuss

   a) how the process of calculating derivative is modified for partial derivatives

   b) how the notations of derivative are modified, and also

   c) how the idea of 'rate of change' evolves.

14.4.1 Partial Derivative of Function with Explicit Representation: Notational & Conceptual Issues

For general discussion, we use the function ***U = f(x, y)***, and for the purpose of specific explanation, we use the following two functions:

i) $z = g(x, y) = 5x^3 y^4$

ii) $v = h(x, y, w) = x^2 \cos(y) / w^2$,

where z and v are *dependent* variables, depending, respectively, on *independent* variable sets {x, y} and {x, y, w}.

We know that, for the function ***U= f (x, y)***,

f_x (x, y) is used to denote the derivative of f(x, y), taking y as a constant, and is defined as

$$\lim_{h\to 0}[\{f(x+h, y) - f(x, y)\}/ h] \qquad \dots (14.4)$$

Alternative notations for $f_x(x, y)$ are

i) $\partial f/ \partial x$,

ii) $\partial U/\partial x$, where U = f (x, y), and sometimes, even the notation

iii) $D_x f$ or $D_x f(x, y)$

Similarly, $f_y(x, y)$ is used to denote the derivative of f(x, y), taking x as a constant, and is defined as

$$lim_{h\to 0}[\{f(x, y+h) - f(x, y)\}/ h] \qquad \dots (14.5)$$

Alternative notations for $f_y(x, y)$ are $\partial f/ \partial y$, $\partial U/\partial y$, where U = f (x, y), and sometimes, even the notation $D_y f(x, y)$.

Similarly, we can define and denote partial derivatives (of first order) for functions of three or more variables.

Example 14.4

i) **For the function** $z = g(x, y) = 5x^3 y^4$, for calculating $g_x(x, y) = \partial g/ \partial x$, we treat y as a constant, and hence,

$g_x(x, y) = 5y^4 \{3 x^2\} = 15 y^4 x^2$.

Similarly, for calculating $g_y(x, y)$, we treat x as a constant, and

$g_y(x, y) = (5x^3)(4 y^3) = 20 x^3 y^3$.

For $v = h(x, y, w) = x^2 \cos(y) / w^2$,

$\partial v/ \partial x = 2x(\cos(y) / w^2)$,

treating w and y *as* constants.

$\partial v/ \partial y = (x^2/ w^2)(-\sin(y)) = -(x^2 \sin(y)/ w^2)$

treating *w* and *x* as constants, and

$\partial v/ \partial w = \{x^2\cos(y)\} \{-2 w^{-3}) = -2 \{x^2\cos(y)/ w^3)$;

treating x and y *as* constants.

14.4.2 Partial Derivative of Function with Implicit Representation

For the purpose of discussing partial derivatives, an** implicit **equation must involve at least THREE variables, e. g.

$$2 x^4 z^3 - 7x^2 y^5 z = x^4 + y^3. \qquad \dots (14.6)$$

And, then

i) Either, **w**e mention explicitly which one of the variables is to be treated as dependent variable, and rest as independent variables,

ii) Or, we mention the partial derivatives required, e. g. $\boldsymbol{\partial y/ \partial x}$ implying y is being treated as dependent variable, and all others as independent variables.

Let us assume we are to calculate *∂z/ ∂x and ∂z/ ∂y, treating implicitly z as dependent variable.*

i) **Differentiating both sides of (14.6) with respect to x, taking y as a constant, we get**

$\mathbf{2}\{(4x^3)z^3 + x^4(3z^2)\,\boldsymbol{\partial z/\partial x}\} - 7y^5\{(2x))z + x^2\boldsymbol{\partial z/\partial x}\} = 4x^3 + 0;$

Keeping terms containing $\partial z/\partial x$ on LHS and shifting the remaining terms on RHS, we get

$2x^4(3z^2)\,\boldsymbol{\partial z/\partial x} - 7y^5(x^2\boldsymbol{\partial z/\partial x}) = \mathbf{4x^3} - 8x^3z^3 + 14y^5xz$

Cancelling out x throughout, and rearranging, we get

$x(6x^2z^2 - 7y^5)\,\boldsymbol{\partial z/\partial x} = \{x^2(4 - 8z^3) + 14y^5z\}$

$\boldsymbol{\partial z/\partial x = \{x^2(4 - 8z^3) + 14y^5z\}/\,x(6x^2z^2 - 7y^5).}$

Similarly,

$\mathbf{2x^4z^3 - 7x^2y^5z = x^4 + y^3.}$ … *(14.7)*

ii) **for $\partial z/\partial y$, differentiating both sides of (d) with respect to y, taking x as a constant, we get**

$(2x^4)(3z^2\,\partial z/\partial y) - (7x^2)\{5y^4z + y^5\,\partial z/\partial y\} = 0 + 3y^2 = 3y^2.$

Keeping terms containing $\partial z/\partial y$ on LHS and shifting the remaining terms on RHS, we get

$\{6x^4z^2 - (7x^2)y^5\}\,\partial z/\partial y = 3y^2 + (7x^2)5y^4z = y^2\{3 + 35x^2y^2z\}$

$\boldsymbol{\partial z/\partial y = (y^2/x^2)\{3 + 35x^2y^2z\}/[\{6x^2z^2 - 7y^5\}.}$

14.4.3 Chain Rule &Partial Derivative when Function is Composite

We, first briefly discuss, in sub-section 14.4.3.1, chain rule for the case of single independent variables only. Next, we extend, in sub-section 14.4.3.2, to the case of two or more independent variables and partial derivatives.

14.4.3.1 Chain Rule for Single Independent Variable

For explaining the ideas involved in this subsection, let us assume that we are given the following function

$\boldsymbol{y = f(x) = (\log(\sin(e^x + 7x^2)),}$

in which we are interested for knowing ***dy/ dx,*** the ***rate of change*** in the value of y at some point w. r.t to the change in independent variable x at that point. Comprehending the function in itself seems to be quite a daunting task. ***The task may be facilitated by exploring it layer by layer as follows***:

At the top level, we consider $y = \log(\sin(e^x + 7x^2))$, as

$y = \log(\ldots)$, with$(\ldots) = (\sin(e^x + 7x^2)) = z$ (say)

Thus, $y = \log z$, with $z = \sin(e^x + 7x^2)$

Next, we consider $z = \sin(\ldots)$, with $(\ldots) = (e^x + 7x^2) = w$(say)

Thus, $z = \sin w$, with $w = (e^x + 7x^2)$

Hence, we have arrived at a representation of y as a chain of simplified functions:

y= log z, so that the derivative of ***y*** w.r.t ***z*** is easily visualised;

z= sin w, so that the derivative of **z** w. r. t ***w*** is easily visualised

w= $e^x + 7x^2$, so that the derivative of **w** w.r.t ***x*** is easily visualised

But how this visualisation helps us in finding the rate of change, i. e. dU/ dx.

The well-known CHAIN RULE helps in simplifying the calculation of complex derivatives like dy/dx. In this particular case, the chain rule states *that*

$$dy/dx = (dy/dz) \times (dz/dw) \times (dw/dx) \qquad \dots (14.8)$$

where,

$(dy/dz) = d(\log z)/dz = 1/z = 1/(\sin w) = 1/(\sin e^x + 7x^2)$

$(dz/dw) = d(\sin w)/dw = \cos w = \cos(e^x + 7x^2)$

$(dw/dx) = d(e^x + 7x^2)/dx = e^x + 14x.$

Substituting these values in (e) above, we get

$dy/dx = \{1/(\sin e^x + 7x^2)\} \times \{\cos(e^x + 7x^2)\} \times \{e^x + 14x\}$

$= (e^x + 14x) \cot(e^x + 7x^2).$

We could have calculated dy/dx without explicit substitutions as follows.

Consider again

$y = \log(\text{Sin}(e^x + 7x^2)),$

At the top level, y is function of some variable, say z

$dy/dx = d(\log(\dots)/d(\dots) \times d(\dots)/dx = [1/(\dots)] \times d(\dots)/dx =$

$[1/(\text{Sin}(e^x + 7x^2))] \times d(\text{Sin}(e^x + 7x^2))/dx,$

For calculation of $d(\text{Sin}(e^x + 7x^2))/dx = [d(\sin(\dots)/d(\dots)] \times d(\dots)/dx = \cos(\dots) \times$

$d((e^x + 7x^2))/dx = \dots$ *and so on.*

In such cases, for faster calculations, we should adopt the latter method without explicit substitutions.

Remarks/ Caution

We may ***think*** that the chain rule

$$dy/dx = (dy/dz) \times (dz/dw) \times (dw/dx) \qquad \dots (14.9)$$

is obtained by simply cancelling out dz (and dw) in the denominator respectively with dz (and dw) in the numerator. **Completely Wrong!**

We must remember that dy/dx is a single symbol, not composed of dy and dx. It is just a matter of chance and prolonged usage that a '/' (slash) is used in it. This slash is not representing fraction. Here, we better consider it as f'(x).

Recall, for y =f(x)

$(dy/dx)_{x=a} = (f'(x))_{x=a} = lim_{h\to 0}[f(a+h) - f(a)]/h.$

Once we take limit, there is no question of fraction.

Coming back to the validity of chain rule, it has been PROVED. Here, we assume it without proof.

14.4.3.2 Chain Rule for Partial Derivatives

The chain rule involving partial derivatives is similar with some modifications explained below. Its applications are explained through a couple of examples.

Case I: When ***w(x, y)*** is function of two variables, and each of **x =x(t) and y =y(t)** is a function of one variable, say t.

In this case, we also have partial derivatives ∂w/ ∂x and ∂w/∂y. But we have only simple derivatives dx/dt and dy/dt, and then follow the chain rule explained in the previous subsection.

Example 14.6

Let $w(x, y) = (2x+3y)^2$, $x= (t+1)^{1/2}$, and $y = (t)^2$.

Then the involved derivatives are ***∂w/ ∂x, ∂w/∂y, dx/dt, dy/dt and also dw/dt***. The last one in view of the fact that

$w(x, y) = (2x+3y)^2 = [2(t+1)^{1/2} + 3t^2]^2$ may be considered as function of single variable t.

Thus,

$$dw/dt = (\partial w/\partial x)(dx/dt) + (\partial w/\partial y)(dy/dt)$$
$$= (\partial (2x+3y)^2/\partial x)(d(t+1)^{1/2}/dt) + (\partial (2x+3y)^2/\partial y)(d(t^2)/dt)$$
$$= [2(2x+3y)]2 \times (½ (t+1)^{-1/2}) + [2(2x+3y)]\, 3 \times (2t) =$$

By substituting values of x and y in terms of t, we may get the final answer in terms of t only

Case II: When ***w (x, y)*** is function of two variables, and each of **x =x(r, s) and y =y(r, s)** is a function of two variables.

In this case, we again have partial derivatives ∂w/ ∂x and ∂w/∂y. Also, as each of x and y is a function of two variables, partial derivatives ∂x/∂r, ∂x/∂s, ∂y/∂r and ∂y/∂sexist. Also, w can be considered as a function of two variables, viz r and s, we have partial derivatives $\partial w/\partial r$, $\partial w/\partial s$ and then follow the chain rule explained in the previous subsection.

Example 14.7

Let $w(x, y) = (2x+3y)^2$, $x= (2r+ s)^{1/2}$ and $y = (r - 2s)^2$.

Then

$\boldsymbol{\partial w/\partial r = \partial w/\partial x\ \partial x/\partial r + \partial w/\partial y\ \partial y/\partial r}$

$= [\{2\,(2x+3y)\}2]\,(1/2)\,(2r + s)^{-1/2} \times 2) + [\{2\,(2x+3y)\}3] \times [\,2\,(r - 2s) \times 1\}, = \ldots$

We may replace variables x and y by variables r and s to get the value of the partial derivative in terms r and s. Similarly,

$\boldsymbol{\partial w/\partial s = \partial w/\partial x\ \partial x/\partial s + \partial w/\partial y\ \partial y/\partial s}$

$= [\{2\,(2x+3y)\}2]\,(1/2)\,(2r + s)^{-1/2} \times 1) + [\{2\,(2x+3y)\}3] \times [\,2\,(r - 2s) \times (-2)], =$
…

We may replace variables x and y by variables r and s. …

14.4.4 Higher Order Partial Derivatives

In the previous subsections 14.4.1 to 14.4.3, we discussed methods of calculating partial derivatives (of first order) of dependent variables as functions of more than one variable. Each of the partial derivatives, so obtained, itself may be considered as some function of the same variables.

For example, ***for the function*** $z = f(x, y) = 5x^3 y^4$,

we obtained $f_x(x, y) = 15\, y^4 x^2$, *and* $f_y(x, y) = 20\, x^3 y^3$.

We may recall the various equivalent notations for f_x *are* $\partial f/\partial x$ and Z_x. Similarly, for Z_y etc.

Now considering $f_x(x, y) = U(x, y) = 15\, y^4 x^2$,

$f_y(x, y) = V(x, y) = 20\, x^3 y^3$,

as new functions, we can differentiate each w.r.t the same independent variables. The derivatives of U (x, y) and V (x, y) will be called ***second order partial derivatives of*** the original function $z = g(x, y)$.

Notation for Higher Order Derivatives:

Now consider

$(z_x)_x = (f_x)_x = \partial U/\partial x = 30y^4x$ *(treating y as a constant)*

$(z_x)_y = (f_x)_y = \partial U/\partial y = 60y^3x^2$ *(treating x as a constant),*

where U itself is $\partial f/\partial x = f_x$. Replacing, in the above equations, we get

$z_{xx} = f_{xx} = \partial/\partial x\,(\partial f/\partial x) = 30y^4x$, and

$z_{xy} = f_{xy} = \partial/\partial y\,(\partial f/\partial x) = 60y^3x^2$

The notations are simplified as

$z_{xx} = f_{xx}=(f_x)_x =\partial^2 f/ \partial x^2 = 30y^4x.$

$z_{xy} = f_{xy} =(f_x)_y= \partial/\partial y\ (\partial f/ \partial x) = \partial^2 f/ \partial y\ \partial x = 60y^3x^2$

Please note the relative orders of x and y as suffixes of f and of ∂. The suffixes for* LATER *variables come on the RIGHT in f, and on the LEFT in ∂.

Similarly, Third Order (partial/ complete) derivatives are derivatives of appropriate Second Order derivatives. And, we may continue so on for still higher order derivatives.

In his respect, we may note:*As we might have observed, the first order partial derivatives Z_x and Z_y of the variable Z have always an* **explicit** *representation— whether the variable z itself is* ***explicitly, implicitly or through composition*** *dependent on the variables x and y.*

Hence, method of finding all higher order derivatives is exactly the same as the one covered under 14.4.1 for EXPLICIT representation.

As calculating higher order derivatives is just a problem of finding appropriate derivatives of the next lower-level ones, calculating Higher Order derivatives is ***NOT a new type*** *of problem. Hence, we will discuss the relevant problems under* ***'Check Your Progress'***

14.4.5 Interpretation of Partial Derivatives

From our knowledge of single variable calculus, we know that derivative of a function (f(x) at a particular point, say x=a, represents the rate of change of the function as the variable x changes around a. Similarly, the partial derivative, ***f_x (x, y)*** represents the rate of change of the function f (x, y) as we change x and hold y fixed; ***and partial derivative f_y (x, y)*** represents the rate of change of f (x, y) as we change y and hold x fixed.

Example 14.8

Determine if ***$f(x, y) = 2x^2/ 3y^3$*** is increasing or decreasing at the point (1, 3),

a) if we allow x to vary and hold y fixed.

b) if we allow y to vary and hold x fixed.

Solution

We know

$f_x(x, y) = \partial (2x^2/ 3y^3)/ \partial x = 4x/(3y^3)$

$f_y(x, y) = \partial (2x^2/ 3y^3)/ \partial y = [(2/3)x^2 (-3)] / y^4 = -2x^2/y^4$

Thus

$(f_x(x, y))_{(1,3)} = (4x/(3y^3))_{(1,3)} = (4/3)(1/27) > 0.$

Hence the function is increasing at (1,3), if x is allowed to change, but y is kept constant

(f_y (x, y)) $_{(1,3)} = (-2x^2/y^4)_{(1, 3)} = -2/81$ **< 0**,

Hence the function is decreasing at (1,3), if y is allowed to change, but x is kept constant.

From the above example, we find that a function, at a point, may be

- ***increasing*** if one, say the first, of the two variables is allowed to change, keeping the other/ second as constant, *and at the same time,*
- ***decreasing*** if the second of the two variables is allowed to change, keeping the first as constant.

14.4.6 Differential/ Total-Differential

We briefly revisit the concept of ***differential,*** also called ***total differential***, which has already been discussed in Unit 11. The concept plays significant role in the disciplines of *Integral Calculus* and *Differential Equations.*

In the case of ***single independent variable***, say x, and some dependent variable y, with y = f (x), each of the notations-cum-terms ***dy*** and ***dx*** is called ***differential***. Further, the two differentials have the following relationship: ***dy = f '(x) dx***

If we are only given f (x), without any dependent variable y, then the differentials are *df* and *dx* and they are related as ***df = f '(x) dx***

In the case of more than one independent variable.

For a function z = f (x, y), the differential **dz** or **df** is given by,

$$dz = f_x\,dx + f_y\,dy \qquad \text{or}$$

$$df = f_x\,dx + f_y\,dy$$

The concept can be extended to functions of three or more variables. For example, for

w = h (x, y, z), a function of three variables, the differential is given by

$$dw = h_x\,dx + h_y\,dy + h_z\,dz, \quad \text{or}$$

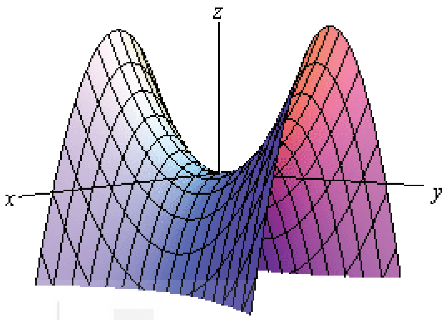
$$dh = h_x\,dx + h_y\,dy + h_z\,dz$$

14.5 DIRECTIONAL DERIVATIVES

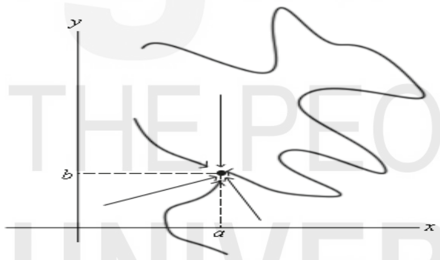
In the previous section, viz. *interpretation of partial derivatives*, we noticed that for a function **z = f(x, y)**, the value of z ***increases*** if x is allowed to change, and y kept fixed, ***and at the same time*** the value of z ***decreases*** if y is allowed to change, and x kept fixed.

Similarly, there may be functions z, in which z ***decreases*** if x is allowed to change, and y kept fixed, ***and at the same time*** the value of z ***decreases*** if y is allowed to change, and x kept fixed.

The later situation may be ***visualized graphically*** *below*, where z= f(x, y) is a surface, in three-dimensional space, and as x increases, with y constant, the value/ surface is moving *downwards*; ***and*** as y increases, with x constant, the value/ surface is moving *upwards.*

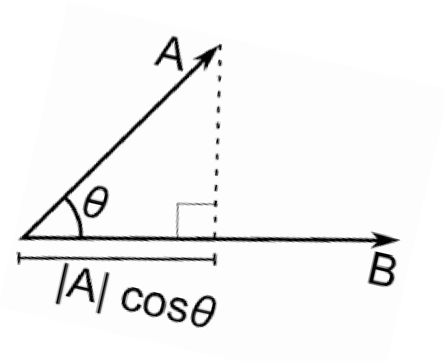


The situation becomes more complex when both x and y are allowed to change simultaneously, specially, as we observed while discussing limits, that a point, in our case, (0, 0), may be approached from ***infinitely*** many directions as shown in the following figure.



The problem of *determining rate of change, which is fundamental issue in calculus,* is made manageable, by fixing the *direction of change*, and then determining the change as we move to the point along that direction. The rate of change, so determined along a direction is called **Directional Derivative.**

More explicitly, by the derivative in the direction **θ,** we mean the rate of change in the value of function f at the point (x, y) with respect to distance $|A| \rightarrow 0$ toward (x, y) along the vector $\vec{A}$that forms the angle **θ** with the positive direction of x-axis.



Formally, ***Directional derivative*** in the direction of the vector $\vec{A}$is given by

$$\mathbf{D\vec{A}f(x,\ y)} = \lim_{|A|\to 0} \frac{f\,(x + |A|\cos\theta,\ \ y + |A|\sin\theta)}{|A|},$$ where, $|A|$ is the length of vector

We state, without proof, the following quite significant result:

For a differentiable function f(x, y), the directional derivative along vector $\vec{A}$is given by

$D_{\vec{A}}f\,(x, y) = f_x \cos\theta + f_y \sin\theta = f_x \cos\theta + f_y \cos(\pi/2 - \theta)$ … … (*)

The unit vector $\vec{A}$is ***specified*** by (cos θ, cos (π/2 – θ)),

where, θ is the angle of direction of vector $\vec{A}$ **w.r.t** positive direction of X-axis.

Recalling the concept of scalar/ inner product of two vectors, from section 6.4 of Unit 6, we can write the expression (*) as ***scalar product*** of two vectors < *fx, fy*> and <cos **θ**,cos **(**π/2 **–θ)** >.

In other words, the directional derivative along unit vector $\vec{A}$is given by

$D_{\vec{A}}f\,(x, y) = f_x \cos\theta + f_y \cos(\pi/2 - \theta) = \langle fx, fy\rangle \cdot \langle \cos\theta, \cos(\pi/2 - \theta)\rangle$… … (*)

The unit vector < *fx, fy*> which occurs frequently in calculus of two variables, is called the **gradient of** f or just **gradient, and is denoted as** ∇f = < *fx, fy*>

Thus from (*), the directional derivative of function f along vector $\vec{A}$ = ∇f. $\vec{A}$

$\vec{A}$, and θ is the angle which vector $\vec{A}$ makes with positive direction of X-axis.

Example 14.9

Find directional derivative for the function

$f(x, y) = x^2 e^{xy} + 3y^2$at the point (0, 2) along the unit vector $\vec{u}$ in the direction of **θ = 2**π/3

The vector $\vec{u}$ **is specified by < cos 2**π/3, sin **2**π/3> = <– ½, (√3)/ 2>

Thus, directional derivative is given as

$$D_{\vec{u}}(x, y) = -\tfrac{1}{2} f_x + ((\sqrt{3})/2)\, f_y$$
$$= -\tfrac{1}{2}\{2x\, e^{xy} + x^2 y\, e^{xy}\} + ((\sqrt{3})/2)\{x^3 e^{xy} + 6y\} = \ldots$$
$$D_{\vec{u}}(0, 2) = -\tfrac{1}{2}\{0\} + ((\sqrt{3})/2)\{0 + 6(2)\} = 6\sqrt{3}$$

Example 14.10

Calculate $D_{\vec{u}}$ **f** (x, y, z) for **f** (x, y, z) = $2xz^2 + 3y^2z^2 - x^2yz$,in the direction of the vector $\vec{v}$**=** <–1, 0, 3>